Competitive Exam Practice 3 — Answer Key

Arrays and Stacks

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Q1 (A). Row-major with 1-based bounds: skip $(i-1)$ full rows of 15 cells, then $(j-1)$ cells, from base 100: $100+(i-1)\times 15+(j-1) = 15i+j+84$. Confirmed against the GATEOverflow online page for GATE CSE 1998 Q2.14 (accepted answer) and ExamSIDE's GATE-1998 array listing (2026-09-09).

Q2 (A). Only scores 51 through 100 are reported — 50 distinct values — so a 50-slot frequency table indexed by $score-51$ is exact and smallest. Confirmed against the BYJUS GATE bank ("Answer (a)") and ExamSIDE's GATE-2005 array listing (2026-09-09).

Q3 3020. 0-based 1D formula $\text{addr}(A[i]) = \text{base} + i \times w = 3000 + 5 \times 4 = 3020$.

Q4 (B). Column-major skips columns: $\text{addr} = \text{base}+((j\times r)+i)\times w = 2000+((3\times 4)+2)\times 4 = 2000+56 = 2056$. (A) 2052 is the row-major trap $((2\times 5)+3)\times 4$; (C) 2044 swaps in $i\times r$; (D) 2068 swaps in $j \times c$.

Q5 (D). The stacks grow toward each other, so no slot is wasted until the tops are adjacent: full iff `top1` = `top2` -1. Fixed-half options (A)/(C) waste space when the stacks are unbalanced. Confirmed against the BYJUS GATE bank ("Answer (d)") and ExamSIDE's GATE-2004 listing (2026-09-09).

Q6 15. Trace the *static* `stkTop`: set size 10; push 5 (`stkTop` $0\to 1$); push 10 ($1\to 2$); first pop returns `A[1]` = 10; second pop returns `A[0]` = 5; printed sum $10+5 = 15$. Cross-checked against ExamSIDE's GATE-2015-Set-2 listing of this question (2026-09-09).

Q7 (A). Trace the index: $-1 \to 0 \to 1 \to 0$ (pop returns 20) $\to 1 \to 2 \to 1$ (pop returns 40); stack holds $[10, 30]$, so `top` = 1 and `peek()` returns 30.

Q8 (B). The last-opened bracket must close first — exactly LIFO, so a stack (push openers, pop on match) decides balance in one $O(n)$ pass.

Q9 (C). Keep the queue's front at the stack top: ENQUEUE(x) = REVERSE, PUSH x, REVERSE (three instructions); DEQUEUE = POP (single). Pairing (B) fails: after one REVERSE+POP the stack is left reversed, so the next PUSH would attach at the wrong end; (A) is refuted by the construction above and (D) is impossible (a single PUSH/POP cannot reverse order). Consistent with GATE's own canonical triple-operation construction from the 2000 descriptive (ENQUEUE = PUSH; DEQUEUE = REVERSE, POP, REVERSE) and listed as GATE 2014 Set 2 on ExamSIDE's stacks-and-queues page (2026-09-09).

Q10 **142.** Left-to-right single pass: $10 + 5 = 15$; $60/6 = 10$; $15 \times 10 = 150$; $150 - 8 = 142$. Expression confirmed on ExamSIDE's GATE-2015-Set-3 page (2026-09-09); arithmetic is exact.